Welcome & Kickoff

Building a Community-Driven Model for Continuous Curriculum Improvement

CMSE Curriculum Development Sprint

https://msucerl.org/cmse-sprint-2026

Session 1: Brainstorming & Team Formation

June 8, 2026

Welcome to the Sprint!

Why are we here?

- **Culture shift:** Moving away from individual, isolated course maintenance toward a collaborative, community-owned approach.
- **Connecting research to practice:** Directly engaging with the Computing Education Research Lab (CERL) to implement evidence-based teaching practices.
- **Valuing your labor:** Ensuring your work is documented as a meaningful indicator of department service and teaching excellence portfolios.

Immediate Goal: Update CMSE curricula for the upcoming academic year.

What We've Done So Far

Our planning up to this point has been entirely driven by your collective input:

- **Written survey:** Collected initial data on departmental "pain points," course-level frustrations, and institutional goals.
- **Two focus group sessions:** Conducted 90-minute deep dives with faculty and graduate TAs.

This generated a rich set of observed themes, student struggles, and programmatic disconnects that will serve as a jumping-off point for our work today.

Goals for this session

Try to move quickly from high-level feedback to hands-on development:

- **Review Key Themes:** Briefly walk through the structural, content-based, and technical categories identified in the focus groups.
- **Broad Brainstorming:** Map these insights into expansive ideas for targeted curriculum improvements.
- **Scope & Triage:** Filter these broad ideas into viable summer sprint projects.
- **Form Teams:** Group up around shared interests and establish our baseline developer roles.

Framing the Focus Group Data

At the highest level, we observed the following:

Classroom Structure & Pedagogical Approaches

Addressing student engagement, the "group rush" mentality, and the widespread impact of Generative AI.

Course Content & Learning Goal Alignment

Targeting content rot, updating stale datasets, and closing persistent gaps in tools like Git.

Technical Implementation & Accessibility

Streamlining local software environments, fixing PDF export blocks, and removing screen-reader barriers.

Project Formation, Step 1: Review the Shared Summary

Take a few minutes to browse the document shared on the sprint website highlighting the exact student friction points across CMSE 201, 202, 314, and 411.

Project Formation, Step 2: Group Up by Mutual Interest

Move to the designated tables or virtual spaces that align closest with your interests:

- **Table 1:** Pedagogy, AI Integration, & Classroom Structure.
- **Table 2:** Course Content, Datasets, & Learning Goals.
- **Table 3:** Tooling, Environments, Grading Automation, & Accessibility.

Project Formation, Step 3: Unconstrained Brainstorming

At your tables, brainstorm curriculum enhancements based on the focus group data.

- **Rule:** Think broadly! Do not worry about scope, labor constraints, or technical limitations yet.
- **Action Item:** Designate a scribe to collect and record all ideas inside our shared document.

> Focus on the ultimate goal: What would the ideal version of this specific module, assignment, or tool look like for students?

Project Formation, Step 4: Triaging to Sprint Scope

Now, work within your groups to filter and shape these ideas into viable summer projects. Use the following criteria:

- **Getting the scope right:** It shouldn't be so simple that it can be completed in an afternoon, but it shouldn't be so massive that it anchors you all summer.
- **Lean into momentum:** Choose something that generates excitement and can establish a highly functional baseline over the next few weeks.
- **Collaborative architecture:** Ensure the project has distinct components so that more than one person can actively contribute.

Team Characteristics & Composition

As we work to form teams, we're aiming to establish small groups of **2 to 4 people** structured with the following ideal profiles:

Domain Expert

At least one person who has recently taught or assisted with the target course in some capacity.

Research Partner

At least one person who works within computing education research to help guide pedagogical choices.

> This cross-functional mix ensures our updates are deeply practical for instructors while remaining pedagogically sound.

Final Step: Group Formation

Facilitating the Match

To ensure everyone lands on a project they care about with a balanced team:

1. Review the consolidated list of triaged projects inside the shared workspace document.
2. Complete the brief preference-ranking form that we'll send out **no later than 2pm**.

We will review the preference data to form our 2-4 person sprint teams and match everyone to a project avenue. Teams will be announced during Session 3.

Thanks!

This was a great first step in building a shared vision for the sprint and connecting your expertise to the work ahead!

Questions or concerns? Reach out to Devin or Sona